

Hensel's Lemma as the p -Adic Pandrosion Iteration

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Abstract

Hensel's lemma lifts approximate roots modulo p to exact roots in $\mathbb{Z}_p$ using the formula $a_{k+1} = a_k - P(a_k)/P'(a_k)$. This is *exactly* the Pandrosion T3 iteration of Paper 1, applied over $\mathbb{Z}_p$ instead of $\mathbb{R}$. The convergence is quadratic in both cases: over $\mathbb{R}$, the error squares; over $\mathbb{Z}_p$, the p -adic precision doubles. We verify by computing $\sqrt{2} \in \mathbb{Z}_7$ to 7^{64} precision, $\sqrt{-1} \in \mathbb{Z}_5$ to 5^{32} precision, and lifting Smyth's root θ_0 of $z^3 - z - 1$ into $\mathbb{Z}_5$.

1 The universal T3 formula

Theorem 1.1 (Hensel = Pandrosion T3 over $\mathbb{Z}_p$). *If $P(a_0) \equiv 0 \pmod{p}$ and $P'(a_0) \not\equiv 0 \pmod{p}$, then*

$$a_{k+1} = a_k - \frac{P(a_k)}{P'(a_k)} \pmod{p^{2^{k+1}}} \quad (1)$$

converges to the unique root $\alpha \in \mathbb{Z}_p$ with $\alpha \equiv a_0 \pmod{p}$.

The convergence table:

Field	Formula	Convergence
$\mathbb{R}$ (Paper 1)	$z \leftarrow z - P(z)/P'(z)$	$ e_{k+1} \leq C e_k ^2$
$\mathbb{C}$ (Paper 2)	same	same (complex)
$\mathbb{Z}_p$ (Paper 95)	same	$v_p(e_{k+1}) \geq 2 v_p(e_k)$

Same formula. Same quadratic convergence. Different field.

2 $\sqrt{2}$ in $\mathbb{Z}_7$

$3^2 = 9 \equiv 2 \pmod{7}$, so $a_0 = 3$. Hensel/T3 lifting:

Step	a_k	Precision	$a_k^2 \equiv 2$
0	3	mod 7	✓
1	10	mod 7^2	✓
2	2166	mod 7^4	✓
3	1 802 916	mod 7^8	✓
4	9.6×10^{12}	mod 7^{16}	✓

Precision doubles at every step: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64$.

3 $\sqrt{-1}$ in $\mathbb{Z}_5$

$2^2 = 4 \equiv -1 \pmod{5}$, so $z^2 + 1$ has a root modulo 5. Hensel lifts to $\sqrt{-1} \in \mathbb{Z}_5$: verified to 5^{32} precision.

This illustrates a key arithmetic phenomenon: $\sqrt{-1}$ exists in $\mathbb{Z}_p$ if and only if $p \equiv 1 \pmod{4}$ (or $p = 2$). Since $5 \equiv 1$, the lifting succeeds.

4 Smyth's root in $\mathbb{Z}_5$

The Smyth polynomial $z^3 - z - 1$ (Paper 83) has $P(2) = 5 \equiv 0 \pmod{5}$ and $P'(2) = 11 \equiv 1 \pmod{5}$. Hensel lifts θ_0 into $\mathbb{Z}_5$ to 5^{32} precision. The plastic number $\theta_0 \approx 1.3247$ over $\mathbb{R}$ and $\theta_0 \equiv 2 \pmod{5}$ in $\mathbb{Z}_5$ are *two incarnations of the same Pandrosion root*.

5 Pandrosion universality

The T3 iteration is a *universal root-finder*: it works over any complete valued field. The Pandrosion identity $F_P \cdot P = P'$ holds over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_p$, $\mathbb{F}_q((t))$, and any other such field. Hensel's lemma is not a separate theorem—it is the p -adic specialization of the Pandrosion dynamics.

References

- [1] K. Hensel, *Über eine neue Begründung*, Jahresber. DMV **6** (1897).
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